

Design hydrostatics report

Designer

Created by

Comment

Filename testmodel5.fbm

Design length	12.000 m	Midship location	6.000 m
Length of buoyancy model	12.000 m	Water density	1.0250
Design beam	3.000 m	Mean shell thickness	0.0000 m
Maximum beam	3.041 m	Appendage coefficient	1.0000
Design draft	0.800 m		

Volume properties		Waterplane properties	
Moulded volume	11.660 m ³	Length on waterline	11.356 m
Total displaced volume	11.660 m ³	Beam on waterline	2.966 m
Displacement	11.951 t	Entrance angle	20.5 degr
Block coefficient	0.4049	Waterplane area	24.30 m ²
Prismatic coefficient	0.5333	Waterplane coefficient	0.6750
Vert. prismatic coefficient	0.5998	Waterplane center of floatation	5.317 m
Wetted surface area	30.02 m ²	Transverse moment of inertia	12.727 m ⁴
Longitudinal center of buoyancy	5.831 m	Longitudinal moment of inertia	174.507 m ⁴
Longitudinal center of buoyancy	-1.492 %		
Vertical center of buoyancy	0.505 m		

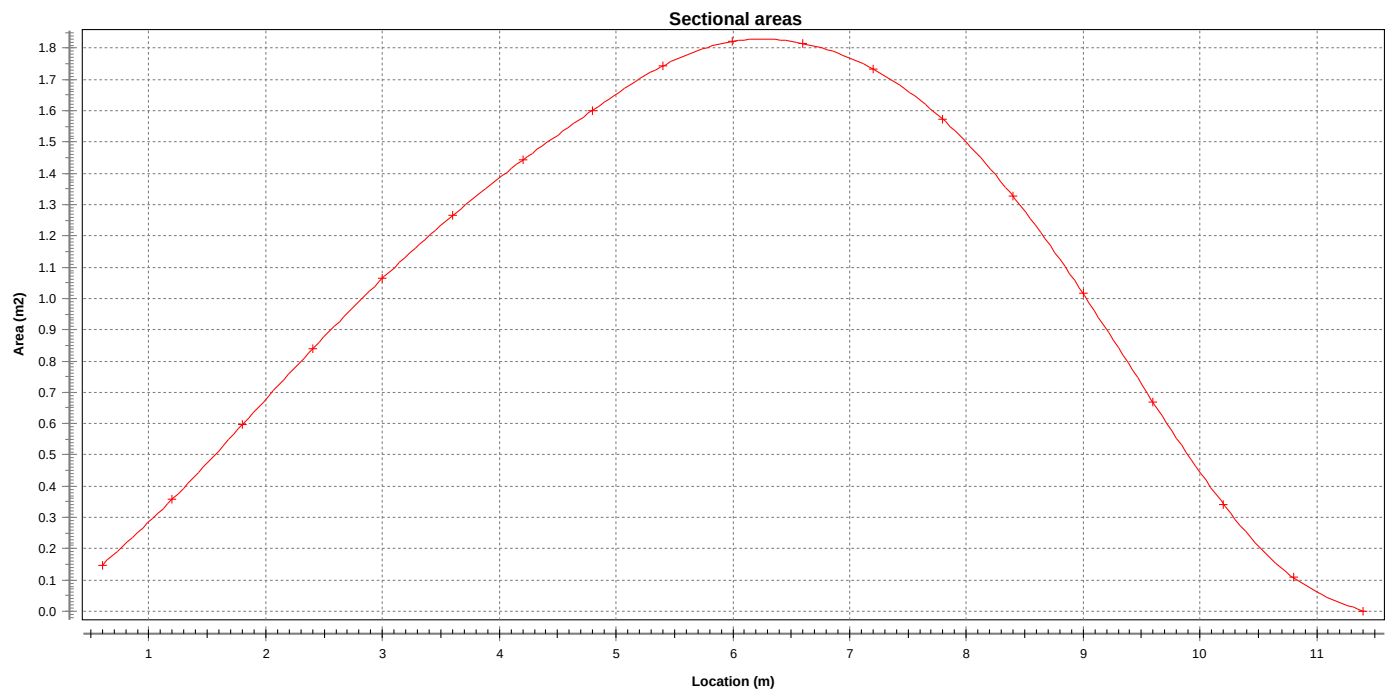
Midship properties		Initial stability	
Midship section area	1.82 m ²	Transverse metacentric height	1.596 m
Midship coefficient	0.7592	Longitudinal metacentric height	15.471 m

Lateral plane	
Lateral area	6.49 m ²
Longitudinal center of lateral resistance	6.233 m
Vertical center of lateral resistance	0.475 m

The following layer properties are calculated for both sides of the ship

Location	Area	Thickness	Weight	LCG	TCG	VCG
	m ²	m	t	m	m	m
Layer 0	83.45	0.000	0.000	5.542	0.000 (CL)	1.126

Sectional areas									
Location	Area	Location	Area	Location	Area	Location	Area	Location	Area
m	m ²	m	m ²	m	m ²	m	m ²	m	m ²
0.600	0.15	3.000	1.06	5.400	1.74	7.800	1.57	10.200	0.34
1.200	0.36	3.600	1.27	6.000	1.82	8.400	1.33	10.800	0.11
1.800	0.60	4.200	1.44	6.600	1.81	9.000	1.02	11.400	0.00
2.400	0.84	4.800	1.60	7.200	1.73	9.600	0.67		



NOTE 1: Draft (and all other vertical heights) is measured from base Z=0.000

NOTE 2: All calculated coefficients based on project length, draft and beam.